Dear Editor,

Please consider our manuscript “Silent but deadly? Physics says seldom, memory says mostly” for publication as a Research Article in *Journal of the Royal Society Interface*.

The belief that a silent release of flatus smells worse than an audible one is recorded in many languages, yet no mechanistic test of it exists. The closest prior work is a conference abstract on the physics of flatulence (Ancalle et al. 2021). We follow one passage through a single chain: chemistry, sound, dispersion, perception and memory. Three results reach beyond the topic.

1. **Routing by the human convective boundary layer.** In simulations of a sofa, a standing pair and a closed lift, the warm air rising along a body takes a slow release next to it and lets a fast one pass, so the momentum of the release, relative to that of the boundary layer, decides which nearby nose it reaches. The same set-up reproduces measured breathing-zone intake from a source at a seated manikin’s groin within a factor of 1.5. This bears on any near-body source, such as respiratory, skin and clothing emissions in shared indoor spaces.

2. **A peak bound for linear transport.** We prove that a slower release of the same substance cannot give a higher peak concentration, in any room or flow. Only the non-linear near field lets it win, by a factor bounded by a velocity ratio, which we call the Silencer number.

3. **Selection across two senses.** With passages as emitted and settings weighted by how often people share them, a bystander finds the silent passage worse in 0.5% of pairs and the audible one in 6.9%; only in a closed lift, for the person behind the releaser, are the two about equal. Yet because a silent passage is noticed only if it is smelled, a remembered silent passage seems the worse one in 88% of comparisons. When an event can be noticed through either of two senses, selection on noticing alone can create a false sensory belief. The explanation makes testable predictions: releasers should hold the belief less firmly than bystanders, and people without a sense of smell should never remember a silent passage.

The work joins fluid dynamics, chemistry, acoustics, sensory science and cognition. We think it suits *Interface* because each part can be read on its own, and because the question shows how a physical and a cognitive explanation can be separated. The headline is checked against the scene weights, the observer models, large-eddy peak factors, mesh refinement and clothing; its direction holds in every case but one adverse pairing of peak factors, and the remembered share stays above 80% in all of them.

Equations and proofs are in appendices, and technical details are in the electronic supplementary material. Code and processed data are openly available at `https://github.com/ccy5123/sbd`; an archived release with a DOI (Zenodo) will accompany the revised manuscript. [**A preprint is posted on engrXiv (DOI . . . ).**] The results have not been reported in the media. The manuscript is not under consideration elsewhere. [**AI-use statement, written by the author.**]

Yours sincerely,

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